Ziad Kheil

zd.kheil@gmail.com
ML Researcher: Biomedical AI and Physics-Informed ML
kheil-z.github.io
github.com/Kheil-Z
linkedin.com/in/ziad-kheil/
Google Scholar

Profile

PhD in deep learning (2025), INSERM / Université de Toulouse. Four years building and training models for **biomedical AI** (CT/MRI, whole-slide pathology, multi-omics) and **physics-informed, self-supervised** methods for 3D/4D imaging.

Skills

Biomedical AI	3D/4D Medical Imaging (CT, MRI), Whole-Slide Pathology, Transcriptomics/Methylation, Deformable Registration, Segmentation (nnU-Net, TotalSegmentator Fine-Tuning)
Physics-Informed ML	Physics-Informed Modeling, Self-Supervised Learning, Representation Learning
Deep Learning	PyTorch, TensorFlow, CUDA, Transformers, Generative Models (GANs, VAEs)
Engineering	Python, Docker, CI/CD, Git
Spoken Languages	English, French, Arabic (Native), Spanish (B2)

Technical Experience

Deep Learning PhD Researcher — **2022 — 2025**
INSERM / IMT / Université de Toulouse — *Toulouse, France*

- Thesis: "Constrained Deep Learning for Deformable Image Registration: When Physics Meets Data."
- Built physics-informed deep learning models embedding physical constraints (sliding, local rigidity) directly into the network, for deformable registration between CT and MRI volumes. Published in Scientific Reports (2026).
- Built a method to recover unobserved anatomical states in 4D-CT sequences from partial temporal context and breathing-amplitude signals. Published at IEEE ISBI (2025).
- Co-authored DRIM, a self-supervised representation-learning method fusing incomplete multimodal data (imaging, transcriptomics, clinical) for glioma outcome prediction. Published at MICCAI (2024).
- Fine-tuned pretrained segmentation models (nnU-Net, TotalSegmentator) and benchmarked architectures (Transformers, U-Nets, GANs, VAEs) on 3D/4D CT and MRI data.
- Built reproducible training pipelines (Docker, CI/CD, Git) across projects, with data sourced and curated directly from hospital systems under institutional ethics review.

Research Internship — Robust Object Detection — **May — Oct. 2022**
IRT Saint-Exupéry / DEEL — *Toulouse, France*

- Integrated Lipschitz-constrained neural networks into object detection pipelines using multi-GPU distributed training, for provable robustness guarantees under adversarial conditions.

Machine Learning Intern — mmWave Point Cloud Classification — **Apr. 2021 — Sep. 2021**
STOFL — *Greater London, UK*

- Built a deep learning classification pipeline for radar point-cloud data (Texas Instruments IWR1443), as part of Imperial College MSc thesis.

Education

PhD in Machine Learning for Healthcare, *INSERM / IMT / Université de Toulouse* — 2022 — 2025
MSc, Machine Learning & Artificial Intelligence, *Imperial College London, UK* — 2020 — 2021
Engineering Degree, Applied Mathematics & Decision Theory, *ISAE-SUPAERO, Toulouse* — 2018 — 2022
Preparatory Classes (MP) — Lycée Michel de Montaigne, Bordeaux* — 2015 — 2018

Projects & Open Source

MONAI
Contributor to the PyTorch medical-imaging framework.

2nd Place — International VisioMel Hackathon (2023)
Predicting melanoma relapse from whole-slide histopathology; €8,000 prize.

Finalist — Owkin & Servier Hackathon (2022)
Spatial transcriptomics for advancing glioblastoma research.

Achievements & Teaching

Supervision
3 MSc students; co-authored the proposal for an ANITI "AI for Health" grant-funded M2 internship (2025).

Teaching
Statistics, AI, deep learning, signal processing, computer vision, Python, R — ISAE-SUPAERO, Université de Toulouse, Toulouse Tech.

Talks
ESTRO 2024 (Glasgow): "Deep-learning deformable image registration for 4DCT/3DMRI label propagation." ISBI 2025 (Houston): "IMITATE: Image Registration with Context for Unknown Time Frame Recovery."

Publications

[1] **Z. Kheil, L. Risser, E. Cohen-Jonathan Moyal, S. Ken.** *"Evaluation of Classical and Deep-Learning Deformable Registration for MR to 4D-CT Contour Mapping in Liver Stereotactic Body Radiotherapy."* Physics and Imaging in Radiation Oncology, 2026.

[2] **Z. Kheil, S. Ken, L. Risser.** *"Incorporating Sliding and Local Rigidity Constraints in Neural-Based Image Registration."* Scientific Reports, 2026.

[3] **P. Deshors, Z. Kheil, L. Ligat, V. Gouazé-Andersson, E. Cohen-Jonathan Moyal.** *"FGFR Inhibition as a New Therapeutic Strategy to Sensitize Glioblastoma Stem Cells to Tumor Treating Fields."* Cell Death Discovery, 2025.

[4] **Z. Kheil, L. Robinet, L. Risser, S. Ken.** *"IMITATE: Image Registration with Context for Unknown Time Frame Recovery."* IEEE International Symposium on Biomedical Imaging (ISBI), 2025.

[5] **L. Robinet, A. Berjaoui, Z. Kheil, E. Cohen-Jonathan Moyal.** *"DRIM: Learning Disentangled Representations from Incomplete Multimodal Healthcare Data."* Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2024.